Problema 4

$T_1 = -4$; $T_2 = 30 \Rightarrow \Delta T = |T_1 - T_2| = |-34| = 34K$

$L = 0,9\ cm = 0,9 \cdot 10^{-2}\ m$

$\Delta t = 1s$

$S = 1\ m^2$

Se alege $K = 0,73\ W/m \cdot K = 0,73\ \frac{J}{s \cdot m \cdot K}$, deoarece este cel mai mic $\Rightarrow$ cel ideal

$$Q = 1 \cdot 1 \cdot \frac{34}{0,9 \cdot 10^{-2}} \cdot 0,73\ \frac{\cancel{s} \cdot \cancel{m^2} \cdot \cancel{K}\ J}{\cancel{m}\ \cancel{s}\ \cancel{m}\ \cancel{K}} = 2002\ J$$

Problema 3

$l = 1,34\ cm = 1,34 \cdot 10^{-2}\ m$; $g = 9,81\ m/s^2$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \sin\theta \quad \Big| \quad \sin\theta = \theta \quad \Rightarrow \quad \frac{d^2\theta}{dt^2} = -\frac{g}{l}\theta$$

$$\theta(t) = A\cos(\omega t + \phi)$$

Obtinem $\omega = \sqrt{\frac{g}{l}}$; $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$

$$T = 2\pi\sqrt{\frac{l}{g}} = 2\pi\sqrt{\frac{1,34 \cdot 10^{-2}}{9,81}} \cong 0,232\ s = 232\ ms$$

Problem 2

$v_0 = 1,7\ m/s$

$b = 1,41\ Kg/m$

$$\frac{dv}{v^2} = -dt \cdot \frac{b}{m} \quad \Big| \int \quad \Rightarrow \quad \int \frac{1}{v^2}dv = -\frac{b}{m}\int dt \quad \Rightarrow$$

$$\Rightarrow -\frac{1}{v} = -\frac{b}{m} \cdot t + C$$

Daca lui t val 0 (mom. initial, $v = v_0$): $-\frac{1}{v_0} = C$

$$\Rightarrow -\frac{1}{v} = -\frac{b}{m}\cdot t - \frac{1}{v_0} \Rightarrow \frac{1}{v} = \frac{b}{m}t + \frac{1}{v_0}$$

v_0 scade cu 90% $\Rightarrow v_0 - 90\%\, v_0 = v_0 - \frac{9v_0}{10} = \frac{v_0}{10}$

în mom respectiv : $\frac{1}{\frac{v_0}{10}} = \frac{b}{m}t + \frac{1}{v_0} \Rightarrow$

$$\Rightarrow \frac{10}{v_0} - \frac{1}{v_0} = \frac{b}{m}t \Rightarrow \frac{9}{v_0} = \frac{b}{m}t \Rightarrow t = \frac{9m}{b\cdot v_0}$$

$$t = \frac{9\cdot 1{,}3}{1{,}41\cdot 1{,}7}\cdot \frac{\cancel{kg}\ \cancel{m}\ s}{\cancel{kg}\ \cancel{m}} = 4{,}88 \text{ secunde} \Rightarrow 4880 \text{ ms}$$

(LA ASTA AM PUS CIFRELE GRESIT ÎN CALCULATOR, DAR CA FORMULE ERA OK :( )